

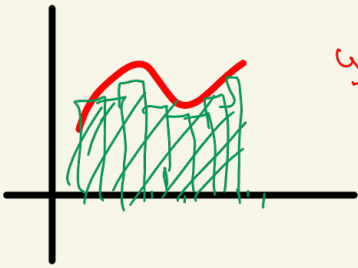

Jan 5, 2026

Part 1: Integrals

Part 2: Application of Integrals

Part 3: Sequences and Series

Definite Integrals



$$y = f(x)$$

If $f(x) > 0$

$$\int_a^b f(x) dx = \text{area below the graph}$$

Conceptual Interpretations =
Continuous accumulation
of quantity.

Infinite Integrals $\int f(x)$

essentially antiderivatives

$x^2 \rightarrow$ antider. $\frac{x^3}{3}$ so

the most general antider.

is $\frac{x^3}{3} + C$.

$\int f(x) dx = F(x) + C$ means

that most general antider.

of $f(x)$

$$F'(x) = f(x) \quad \int_0^1 x^2 dx = \frac{1}{3}$$

$$\int x^2 dx = \frac{x^3}{3} + C$$

FTC: Fundamental Theory of Calculus

$$\text{FTC: } \int_0^1 x^2 dx = \left[\frac{x^3}{3} \right]_0^1 = \frac{1}{3} - \frac{0}{3} = \frac{1}{3}$$

this always works

January 7, 2026

$$(f(x)g(x))' = f'(x)g(x) + f(x)g'(x)$$

$$\int (f'(x)g(x) + f(x)g'(x)) dx = f(x)g(x) + C$$

$$\int f'(x)g(x) dx + \int f(x)g'(x) = f(x)g(x) + C$$

$$\int f(x)g'(x) = f(x)g(x) - \int f'(x)g(x) dx$$

Integration by parts

$$\int x^2 dx - 7 = \frac{x^3}{3} + (C-7) = \int x^2 dx$$

1)

$$\int x \cdot \cos x dx$$

Integration by
parts

Solution:

$$\begin{aligned}
 \int x \cdot \cos x \, dx &= \int x \cdot (\sin x)' \, dx = \\
 &= x \cdot \sin x - \int x' \sin x \, dx = x \cdot \sin x - \\
 &- \int \sin x \, dx = x \cdot \sin x + \cos x + C
 \end{aligned}$$

$$\begin{aligned}
 2) \int 2^x \cdot x \, dx &= \int \left(\frac{2^x}{\ln 2} \right)' \cdot x \, dx = \\
 &= \frac{2^x}{\ln 2} \cdot x - \int \frac{2^x}{\ln 2} \cdot 1 \, dx =
 \end{aligned}$$

$$= \frac{2^x \cdot x}{\ln 2} - \frac{1}{\ln 2} \int 2^x \, dx \rightarrow \frac{2^x}{\ln 2}$$

$$= \frac{2^x \cdot x}{\ln 2} - \frac{2^x}{(\ln 2)^2}$$

$$\begin{aligned}
 3) \int x^2 e^x dx &= \int x^2 \cdot (e^x)' dx = \\
 &= x^2 e^x - \int (x^2)' \cdot e^x dx = x^2 e^x - 2 \int \\
 &= 2 \int \underbrace{x \cdot e^x}_{\text{blue box}} dx = x^2 e^x - 2 \int x \cdot (e^x)' dx \\
 &= x^2 \cdot e^x - 2 \left(x \cdot e^x - \int \boxed{x'}^{\uparrow 1} e^x dx \right) = \\
 &= x^2 \cdot e^x - 2 \cdot x \cdot e^x + 2 \boxed{\int e^x dx} e^x \\
 &= \boxed{x^2 \cdot e^x - 2x \cdot e^x + 2 \cdot e^x + C}
 \end{aligned}$$

$$\begin{aligned}
 4) \int \ln x dx &= \int \boxed{1}^{x'} \cdot \ln x dx = \\
 &= \int x' \ln x dx = x \ln x - \int x (\ln x)' dx \\
 &= x \ln x - \int x \cdot \frac{1}{x} dx = x \ln x - \int 1 dx
 \end{aligned}$$

$$= \boxed{x \cdot \ln x - x + C}$$

$$5) \int \cos^2 x \, dx = \int \cos x \cdot (\sin x)' \, dx$$

$$= \cos x \cdot \sin x - \int (\cos x)' \sin x \, dx$$

$$= \cos x \cdot \sin x + \boxed{\int \sin^2 x \, dx}$$

$$= \cos x \cdot \sin x + \int (1 - \cos^2 x) \, dx$$

$$= \cos x \cdot \sin x + \int 1 \, dx - \int \cos^2 x \, dx$$

$$= \cos x \cdot \sin x + x - \int \cos^2 x \, dx$$

Solve for $\int \cos^2 x \, dx$:

$$2 \int \cos^2 x \, dx = \frac{1}{2} (\cos x \sin x + x) + C$$

$$u = x^2$$

$$u = g(x)$$

$$\frac{du}{dx} = 2x$$

$$g'(x)$$

$$du = 2x \, dx$$

$$g'(x) \, dx$$

$$\int \underbrace{f(x)}_u \underbrace{g'(x)}_{dv} \, dx = \underbrace{f(x)}_u \underbrace{g(x)}_v - \int \underbrace{g(x)}_v \underbrace{f'(x)}_{du}$$

$$\int u \, dv = uv - \int v \, du$$

1)

$$\int \underbrace{x}_u \cdot \underbrace{\cos x \, dx}_{dv} :$$

$$u = x \quad v = \sin x$$

$$dv = \cos x \cdot dx$$

$$du = dx$$

$$\int \underbrace{x}_u \cdot \underbrace{\cos x \, dx}_{dv} = \int u \, dv = uv - \int v \, du + C$$

$$= x \cdot \sin x - \int \sin x \, dx = x \cdot \sin x - \cos x + C$$

January 12, 2026

Substitution Rule (aka "u-sub")

$$\int (f(x) + g(x)) \, dx = \int f(x) \, dx + \int g(x) \, dx$$

$\int f(x) g(x) \, dx \rightarrow$ integration by parts

$$f(x) g(x) - \int f'(x) g(x) \, dx$$

* often serves the purpose

$\int f(g(x)) \, dx \quad \text{X} \rightarrow$ no such thing

$\int f(g(x)) g'(x) \, dx \rightarrow$ there is a formula for this, substitution rule

$$\int f(g(x))g'(x) dx = \int \underbrace{F'(g(x))g'(x)}_{(F(g(x)))'} dx$$

$$= \int (F(g(x)))' dx = F(g(x)) + C \quad \text{Let } u = g(x)$$

$$= F(u) + C = \int f(u) du \quad \checkmark$$

$$\text{let } u = x^2$$

$$\frac{du}{dx} = 2x$$

$$du = 2x dx$$

$$u = g(x)$$

$$du = g'(x) dx$$

$$\frac{du}{dx} = g'(x)$$

$$\int \underbrace{f(g(x))}_u \underbrace{g'(x)}_{du} dx = \int f(u) du$$

Practice

1.

$$\int x \cdot \cos(x^2) dx$$

$$\text{let } u = x^2$$

$$du = 2x dx \rightarrow \frac{du}{2}$$

$$\int x \cdot \cos(x^2) dx = \int \cos(u) \cdot \boxed{\frac{du}{2}} =$$

$$\frac{1}{2} \int \cos(u) du = \frac{1}{2} \sin(u) + C =$$

$$\frac{1}{2} \sin(x^2) + C$$

2.

$$\int x^2 \sqrt[3]{4x^3+3} dx$$

$$\text{let } u = 4x^3+3$$

$$du = 12x^2 dx \mid x^2 dx = \frac{du}{12}$$

$$\int x^2 \sqrt[3]{4x^3+3} = \int \sqrt[3]{u} \frac{du}{12} = \frac{1}{2} \int \sqrt[3]{u} du$$

$$= \frac{1}{12} \int u^{\frac{1}{3}} du = \frac{1}{12} \frac{u^{\frac{1}{3}+1}}{\frac{1}{3}+1} + C =$$

$$= \frac{u^{\frac{4}{3}}}{16} + C = \boxed{\frac{(4x^3+3)^{\frac{4}{3}}}{16} + C}$$

3.

$$\int \sin(10x) dx \quad \begin{array}{l} \text{let } u = 10x \\ du = 10 dx \\ dx = \frac{du}{10} \end{array}$$

$$\int \sin(10x) dx = \int \sin(u) \frac{du}{10} = \frac{1}{10} \int \sin(u) du$$

$$= -\frac{1}{10} \cos(u) + C = \boxed{-\frac{1}{10} \cos(10x) + C}$$

Useful shortcut:

If $\int f(x) dx = F(x) + C$, then

$$\cdot \int f(x+a) dx = F(x+a) + C$$

$$\int \cos(x+7) dx = \sin(x+7) + C$$

$$\int f(a \cdot x) dx = \frac{F(a \cdot x)}{a} + C$$

$$\int \cos(7 \cdot x) dx = \frac{\sin(7x)}{7} + C$$

$$5. \int \frac{x^3}{1+x^4} dx$$

Solution:

$$\text{Let } u = 1+x^4$$

$$du = 4x^3 dx \quad x^3 dx = \frac{du}{4}$$

$$\int \frac{x^3}{1+x^4} dx = \int \frac{du/4}{u} = \frac{1}{4} \int \frac{1}{u} du =$$

$$= \frac{1}{4} \ln|x| + C = \boxed{\frac{1}{4} \ln|1+x^4| + C}$$

$$6. \int \frac{x}{1+x^4} dx$$

$$\int \frac{x}{1+x^4} dx = \int \frac{x}{1+(x^2)^2} dx$$

$$\text{Let } u = x^2$$

$$du = 2x dx \quad x dx = \frac{du}{2}$$

$$\int \frac{1}{1+x^4} x dx = \int \frac{1}{1+u^2} \frac{du}{2} =$$

$$= \frac{1}{2} \int \frac{1}{1+u^2} du = \frac{1}{2} \tan^{-1}(u) + c$$

$$= \boxed{\frac{1}{2} \tan^{-1}(x^2) + c}$$

$$7. \int \frac{1}{x^2 + 14x + 130} dx$$

$$\text{Solution: } x^2 + 14x + 130 = (\quad)^2 + c$$

$$(a+b)^2 = a^2 + 2ab + b^2$$

$$x^2 + 14x + 130 = (x^2 + 2 \cdot x \cdot 7 + 7^2) + 7^2 + 130$$

$$= (x+7)^2 + 81 \rightarrow \text{Completing the square}$$

$$\int \frac{1}{(x+7)^2 + 81} dx \quad \text{Let } u = x+7$$

$$du = dx$$

$$\int \frac{1}{u^2 + 81} du = \int \frac{1}{u^2 + 9^2} du =$$

$$= \frac{1}{9} \tan^{-1}\left(\frac{u}{9}\right) + C = \boxed{\frac{1}{9} \tan^{-1}\left(\frac{x+7}{9}\right) + C}$$

11

$$\int \frac{\sin x \cdot \cos x}{1 + \sin^2 x} dx$$

$$\text{Let } u = \sin x \quad du = \cos x \, dx$$

$$\int \frac{\sin x \cos x}{1 + \sin^2 x} dx = \int \frac{u}{1+u^2} du$$

$$\text{Let } v = 1+u^2$$

$$dv = 2u du$$

$$u \cdot du = \frac{dv}{2}$$

$$\int \frac{1}{v} \frac{dv}{2} = \frac{1}{2} \int \frac{1}{v} dv = \frac{1}{2} \ln|v| + c$$

$$\frac{1}{2} \ln(1 + \sin^2 x) + c$$

Alternative:

$$\text{Let } u = 1 + \sin^2 x \quad du = 2 \sin \cdot \cos x$$

$$\begin{aligned} \int \frac{\sin x \cdot \cos x}{1 + \sin^2 x} &= \frac{1}{2} \int \frac{1}{u} du = \frac{1}{2} \ln|u| + c \\ &= \frac{1}{2} \ln(1 + \sin^2 x) + c \end{aligned}$$

$$12. \int x^5 \cdot \cos(x^3) = \int x^2 \cdot x^3 \cdot \cos(x^3) dx$$

Solution: Let $u = x^3$
 $du = 3x^2 dx$

$$\int \underbrace{x^2 x^3 \cos(x^3)}_{u \cdot \cos u} dx = \int u \cdot \cos(u) \frac{du}{3} =$$

$$= \frac{1}{3} \int u \cdot \cos u \, du = \frac{1}{3} \int u \cdot (\sin' u) \, du =$$

$$= \frac{1}{3} \left[u \cdot \sin(u) - \int u' \sin u \, du \right] = \frac{1}{3} ($$

$$= \frac{1}{3} (u \cdot \sin u - \int \sin u \, du) = \frac{1}{3} (u \cdot \sin u + \cos u) + C = \frac{1}{3} (x^3 \cdot \sin x^3 + \cos(x^3)) + C$$

January 14, 2026

Trigonometric Integrals

1. Integrals of form $\int \sin^m x \cos^n x dx$
2. Integrals of form $\int \tan^m x \sec^n x dx$
3. Integrals of form $\int \sin(mx) \sin(nx) dx$
... etc

$$1 \int \sin^m x \cdot \cos^n x dx$$

$$\int \cos^3 x dx = \int (1 - \sin^2 x) \cos x dx =$$

$$\cos^3 x = \cos x \cdot \cos^2 x = \cos x (1 - \sin^2 x)$$

$$\int \underbrace{(1 - \sin^2 x)}_{1-u^2} \underbrace{\cos x dx}_{du} = \int (1 - u^2) du$$

$$\text{let } u = \sin x$$

$$du = \cos x$$

$$\int \sin^m x \cdot \cos^n x \, dx$$

Case 1: n is odd

$$n = 2k + 1$$

$$\int \sin^m x \cdot \cos^{2k+1} x \, dx = \int \sin^m x \cdot (\cos^2 x)^k \cos x \, dx$$

$$= \int \sin^m x \cdot (1 - \sin^2 x)^k \cos x \, dx$$

Case 2: m is odd ($m = 2k + 1$)

$$\int \sin^{2k+1} x \cdot \cos^n x \, dx = \int \sin^{2k} x \cdot \sin x \cdot \cos^n x \, dx$$

$$= \int (1 - \cos^2 x)^k \cdot \cos^n x \cdot \sin x \, dx \quad u = \cos x$$

Case 3: m & n odd, do any

Ex.

$$\int \sin^6 x \cdot \cos^5 x \, dx = \int \sin^6 x \cdot \cos^4 x \cdot \cos x \, dx$$

$$= \int \sin^6 x \cdot (\cos^2 x)^2 \cos x \, dx \quad \begin{array}{l} \text{let } u = \sin x \\ du = \cos x \, dx \end{array}$$

$$= \int \underbrace{\sin^6 x \cdot (1 - \sin^2 x)^2}_{u^6 \cdot (1 - u^2)^2} \underbrace{\cos x \, dx}_{du}$$

$$= \int u^6 (1 - u^2)^2 \, du = \int u^6 (1 - 2u^2 + u^4) \, du$$

$$= \int (u^6 - 2u^8 + u^{10}) \, du = \frac{u^7}{7} - 2 \frac{u^9}{9} + \frac{u^{11}}{11} + C$$

$$= \frac{\sin^7 x}{7} - \frac{2 \sin^9 x}{9} + \frac{\sin^{11} x}{11} + C$$

Ex. $\int \sin^4 x \cdot \cos^6 x \, dx = \int (\sin^2 x)^2 (\cos^2 x)^3 \, dx$

$$= \int \left(\frac{1 - \cos(2x)}{2} \right)^2 \left(\frac{1 + \cos(2x)}{2} \right)^3 \, dx$$

Ex. $\int \cos^2 x \, dx$

Solution:

$$\int \cos^2 x \, dx = \int \frac{1 + \cos(2x)}{2} \, dx =$$

$$\frac{1}{2} \left(x + \int \cos(2x) \, dx \right) = \frac{x}{2} + \frac{1}{2} \cdot \frac{\sin(2x)}{2} + C$$

$$\int \cos(2x) = \frac{\sin(2x)}{2}$$

Ex. $\int \cos^2 x \cdot \sin^2 x \, dx = \int \frac{1 - \cos(2x)}{2} \cdot \frac{1 + \cos(2x)}{2} \, dx$

$$= \frac{1}{4} \int (1 - \cos(2x))(1 + \cos(2x)) \, dx =$$

$$= \frac{1}{4} \int (1 - \cos^2(2x)) \, dx = \frac{1}{4} x - \frac{1}{4} \int \cos^2(2x)$$

$$= \frac{x}{4} - \frac{1}{4} \int \frac{1 + \cos(4x)}{2} \, dx$$

$$= \frac{x}{4} - \frac{x}{4} - \frac{1}{8} \int \cos(4x) dx$$

$$= \frac{x}{8} - \frac{1}{8} \cdot \frac{\sin(4x)}{4} + C = \boxed{\frac{x}{8} - \frac{\sin(4x)}{32} + C}$$

$$\int \tan^{\overset{\text{odd}}{\textcircled{m}}} x \cdot \sec^{\overset{\text{even}}{\textcircled{n}}} x dx$$

1. If n is even, $n = 2k$

$$\int \tan^m x \cdot \sec^{2k} x dx = \int \tan^m x \cdot \underbrace{\sec^{2k-2} x}_{(\sec^2 x)^{k-1}} \cdot \sec^2 x$$

$$= \int \underbrace{\tan^m x (1 + \tan^2 x)^{k-1}}_{\text{terms of } u} \underbrace{\sec^2 x}_{du} dx$$

2. If $m = 2k+1$

$$\int \tan^{10} x \cdot \sec^6 x dx = \int \tan^{10} x \cdot \sec^4 x \cdot \sec^2 x$$

$$= \int \tan^{10} x (\sec^2 x)^2 \cdot \sec^2 x dx$$

$$= \int \tan^{10} x (1 + \tan^2 x)^2 \cdot \sec^2 x \, dx$$

$$\text{let } u = \tan x$$

$$du = \sec^2 x \, dx$$

$$= \int u^{10} (1 + u^2)^2 \, dx = \int u^{10} (1 + 2u^2 + u^4) \, du$$

$$= \int (u^{10} + 2u^{12} + u^{14}) \, du = \frac{u^{11}}{11} + 2 \frac{u^{13}}{13} + \frac{u^{15}}{15} + C$$

$$= \frac{\tan^{11} x}{11} + \frac{2 \tan^{13} x}{13} + \frac{\tan^{15} x}{15} + C$$

$$\int \cot^3 x \cdot \csc^3 x \, dx$$

Solution:

$$\int \cot^2 x \cdot \csc^2 x (\cot x \cdot \csc x) \, dx =$$

$$\int (\csc^2 x - 1) \csc^2 x \cot x \cdot \csc x \, dx$$

$$\text{let } u = \csc x$$

$$du = -\cot x \csc x \, dx$$

$$= -\int (u^2 - 1) u^2 \, du = -\int (u^4 - u^2) \, du =$$

$$= \int (u^2 - u^4) \, du = \frac{u^3}{3} - \frac{u^5}{5} + C$$

$$= \frac{\csc^3 x}{3} - \frac{\csc^5 x}{5} + C$$

$$1 \int \cos(7x) \cdot \cos(4x) \, dx =$$

$$= \int \frac{1}{2} (\cos(7-4x) + \cos(7x+4x)) \, dx$$

$$= \frac{1}{2} \int (\cos(3x) + \cos(11x)) \, dx$$

$$= \frac{1}{2} \left(\frac{\sin(3x)}{3} + \frac{\sin(11x)}{11} \right) + C$$

Tutorial

January 16, 2026

$$1. \int x^2 \cdot \tan(x^3 + 1) dx$$

Solution Let $u = x^3 + 1$

$$du = 3x^2 dx \rightarrow \frac{du}{3} = x^2 dx$$

$$\int x^2 \cdot \tan(x^3 + 1) dx$$

$$= \int \frac{du}{3} \cdot \tan(u) = \frac{1}{3} \int \tan(u) du =$$

$$= \frac{1}{3} \ln|\sec(u)| + C = \frac{1}{3} \ln|\sec(x^3 + 1)| + C$$

$$2. \int \frac{x^9}{4+x^{10}} dx$$

$$\text{Let } u = 4+x^{10}$$

$$du = 10x^9$$

$$x^9 = \frac{du}{10}$$

$$\int \frac{x^9}{4+x^{10}} = \frac{1}{10} \int \frac{1}{u} \cdot du = \frac{1}{10} \ln|u| + C$$

$$= \frac{1}{10} \ln|4+x^{10}| + C$$

$$3. \int \frac{x^4}{4+x^{10}} dx = \int \frac{x^4}{4+(x^5)^2} dx$$

$$\text{Let } u = x^5$$

$$du = 5x^4 dx$$

$$x^4 dx = \frac{du}{5}$$

$$= \frac{1}{5} \int \frac{1}{4+u^2} du = \frac{1}{5} \int \frac{1}{2^2+u^2} du =$$

$$= \frac{1}{5} \cdot \frac{1}{2} \tan^{-1} \left(\frac{u}{2} \right) + C =$$

$$= \frac{1}{5} \cdot \frac{1}{2} \cdot \tan^{-1} \left(\frac{x^5}{2} \right) + C$$

$$4. \int \frac{1}{\sqrt{21+4x-x^2}}$$

Solution

$$21+4x-x^2 = 21-(x^2-4x) =$$

$$= \underline{21} - \underbrace{(x^2 - 2 \cdot x \cdot 2 + 2^2)}_{(x-2)^2} - \underline{2^2} = 25 - (x-2)^2$$

$$\int \frac{1}{\sqrt{25 - (x-2)^2}} dx$$

$$\text{Let } u = x-2 \\ du = dx$$

$$= \int \frac{1}{\sqrt{25-u^2}} du$$

$$\int \frac{1}{\sqrt{a^2-x^2}} = \sin^{-1}\left(\frac{x}{a}\right) + C$$

$$= \int \frac{1}{\sqrt{5^2-u^2}} du = \sin^{-1}\left(\frac{u}{5}\right) + C$$

$$= \boxed{\sin^{-1}\left(\frac{x-2}{5}\right) + C}$$

$$6. \int x^5 \sqrt{4+x^2} dx$$

Solution

$$\text{Let } u = 4+x^2$$

$$du = 2x dx$$

$$\boxed{x dx = du/2}$$

$$\int x^5 \underbrace{\sqrt{4+x^2}}_{u^2} dx = \int \underbrace{x \cdot (x^2)^2}_{\text{red circle}} \underbrace{\sqrt{4+x^2}}_{\text{red circle}} dx \quad \frac{du}{2}$$

$$= \frac{1}{2} \int (u-4)^2 \sqrt{u} \, du \rightarrow \text{brute force}$$

$$= \frac{1}{2} \int (u^2 - 8u + 16) u^{\frac{1}{2}} \, du =$$

$$= \frac{1}{2} \int (u^{\frac{5}{2}} - 8u^{\frac{3}{2}} + 16u^{\frac{1}{2}}) \, du =$$

$$= \frac{1}{2} \left(\frac{u^{\frac{7}{2}}}{\frac{7}{2}} - 8 \frac{u^{\frac{5}{2}}}{\frac{5}{2}} + \frac{16u^{\frac{3}{2}}}{\frac{3}{2}} \right) + C =$$

$$= \frac{(x^2+4)^{\frac{7}{2}}}{7} - 8 \frac{(x^2+4)^{\frac{5}{2}}}{5} + 16 \frac{(x^2+4)^{\frac{3}{2}}}{3} + C$$

$$7. \int \frac{1}{1+e^{-x}} \, dx = \int \frac{1}{1+\frac{1}{e^x}} \, dx =$$

$$= \int \frac{e^x}{e^x+1} \, dx =$$

$$\text{let } u = e^x + 1$$

$$du = e^x \, dx$$

$$= \int \frac{1}{u} du = \ln|u| + C = \ln(e^x + 1) + C$$

TRIG INTEGRALS

$$9. \int \tan^{\textcircled{3}} x \cdot \sec^1 x \, dx =$$

odd ✓

$$= \int \tan^2 x \cdot \tan x \cdot \sec x \, dx =$$

$$= \int (\sec^2 - 1) \tan x \cdot \sec x \, dx$$

$$\text{Let } u = \sec x$$

$$du = \tan x \cdot \sec x \, dx$$

$$= \int (u^2 - 1) du = \frac{u^3}{3} - u + C =$$

$$\frac{\sec^3}{3} - \sec x + C$$

$$\begin{aligned}
& \int \cos^4 x \, dx = \int (\cos^2 x)^2 \, dx \\
&= \int \left(\frac{1 + \cos(2x)}{2} \right)^2 \, dx \\
&= \frac{1}{4} \int (1 + \cos(2x))^2 \, dx = \frac{1}{4} \int 1 + 2 \cdot \cos(2x) + \cos^2(2x) \, dx \\
&= \frac{x}{4} + \frac{1}{2} \int \cos(2x) \, dx + \frac{1}{4} \int \cos^2(2x) \, dx \\
&= \frac{x}{4} + \frac{1}{2} \frac{\sin(2x)}{2} + \frac{1}{4} \int \frac{1 + \cos(4x)}{2} \, dx \\
&= \frac{x}{4} + \frac{\sin(2x)}{4} + \frac{1}{8} \int 1 + \cos(4x) \, dx = \\
&= \frac{x}{4} + \frac{\sin(2x)}{4} + \frac{x}{8} + \frac{1}{8} \frac{\sin(4x)}{4} + C \\
&= \frac{3x}{8} + \frac{\sin(2x)}{4} + \frac{\sin(4x)}{32} + C
\end{aligned}$$

January 19, 2026

Trigonometric Substitution

$$\int x^2 \sqrt{16 - x^2} dx = \int 16 \sin^2 \theta \sqrt{16 - 16 \sin^2 \theta} \cdot 4 \cos \theta d\theta$$

$\Downarrow$
 $\sqrt{4^2 - x^2}$

$$= 4 \sqrt{1 - \sin^2 \theta} = 4 \sqrt{\cos^2 \theta}$$

$$x = 4 \sin \theta$$

$$= 4 \cos \theta$$

$$dx = 4 \cos \theta d\theta$$

$$= 256 \int \sin^2 \theta \cos^2 \theta d\theta$$

trig integral

1. For $\sqrt{a^2 - x^2}$

$x = a \cdot \sin \theta$, choose θ such that
this happens

Is there even such θ ?

$$a^2 - x^2 \geq 0 \quad x^2 \leq a^2 \quad -a \leq x \leq a$$

yes, because $-a \leq x \leq a$

$$\begin{aligned}\sqrt{a^2 - x^2} &= \sqrt{a^2 - a^2 \sin^2 \theta} = a \sqrt{1 - \sin^2 \theta} \\ &= a \sqrt{\cos^2 \theta} = a |\cos \theta| = a \cdot \cos \theta\end{aligned}$$

2.

For $\sqrt{a^2 + x^2}$ no restrictions

$$\text{let } x = a \cdot \tan \theta$$

$$\begin{aligned}\sqrt{a^2 + x^2} &= \sqrt{a^2 + a^2 \tan^2 \theta} = a \cdot \sqrt{\sec^2 \theta} \\ &= a |\sec \theta| = a \cdot \sec \theta\end{aligned}$$

3.

$$\sqrt{x^2 - a^2}$$

$$x^2 - a^2 \geq 0$$

$$x^2 \geq a^2$$

$$\underline{x \geq a}$$

or

$$\underline{x \leq -a}$$

Ex 1.

$$\int x \sqrt{16-x^2} dx = \int x \sqrt{4^2-x^2} dx$$

$$\text{let } x = 4 \sin \theta \left(-\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2} \right)$$

$$dx = 4 \cos \theta d\theta$$

$$\text{Then } \sqrt{16-x^2} = \sqrt{16-16 \sin^2 \theta} =$$

$$= \sqrt{16(1-\sin^2 \theta)} = 4 \sqrt{\cos^2 \theta} = 4 |\cos \theta|$$

$$= 4 \cos \theta$$

$$\text{because } -\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$$

$$\int x \sqrt{16-x^2} dx = 4 \int 4 \sin \theta \cdot 4 \cos \theta \cdot 4 \cos \theta$$

$$= 64 \int \sin \theta \cos^2 \theta d\theta$$

$$\text{let } u = \cos \theta$$

$$du = -\sin \theta d\theta$$

$$= 64 \int \cos^2 \theta \underbrace{\sin \theta}_{-d\theta} d\theta$$

$$= -64 \int u^2 du = \boxed{-\frac{64}{3} u^3 + C}$$

$$= -\frac{64}{3} \cdot \cos^3 \theta + C$$

$$\sin \theta = \frac{x}{4}$$

$$\theta = \sin^{-1}\left(\frac{x}{4}\right)$$

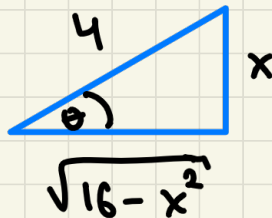
because $-\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$

$$= \boxed{-\frac{64}{3} \cos^3\left(\sin^{-1}\left(\frac{x}{4}\right)\right) + C}$$

If $x = 4 \sin \theta$, $-\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$

compute $\cos \theta$ in terms of x

$$\sin \theta = \frac{x}{4}$$



$$\cos \theta = \frac{\sqrt{16 - x^2}}{4}$$

Conclusion

$$\int x \sqrt{16-x^2} dx = -\frac{64}{3} \cos^3 \theta + C$$

$$= -\frac{64}{3} \left(\frac{\sqrt{16-x^2}}{4} \right)^3 + C$$

$$= -\frac{(\sqrt{16-x^2})^3}{3} + C$$

2.

$$\int \frac{x^3}{\sqrt{x^2+9}} dx$$

$x^2 + 3^2$

Solution

let $x = 3 \tan \theta$

$dx = 3 \sec^2 \theta d\theta$

Then $\sqrt{x^2+9} = \sqrt{9 \tan^2 \theta + 9} = 3 \sqrt{\tan^2 \theta + 1}$

$$= 3 \sqrt{\sec^2 \theta} = 3 |\sec \theta| = 3 \sec \theta$$

$$\int \frac{x^3}{\sqrt{x^2+9}} = \int \frac{27 \tan^3 \theta}{3 \sec \theta} \cdot 3 \sec^2 \theta d\theta$$

$$= 27 \int \tan^3 \theta \sec \theta d\theta$$

$$= 27 \int \tan^3 \theta \cdot \sec \theta \, d\theta$$

$$= 27 \int \tan^2 \theta \cdot \tan \theta \cdot \sec \theta \, d\theta$$

$$= 27 \int (\sec^2 - 1) \tan \theta \cdot \sec \theta \, d\theta$$

$$\text{let } u = \sec \theta$$

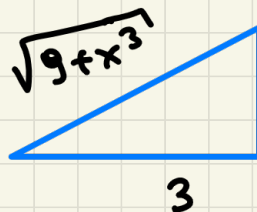
$$du = \tan \theta \sec \theta \, d\theta$$

$$= 27 \int (u^2 - 1) \, du = 27 \left(\frac{u^3}{3} - u \right) + C$$

$$= 9u^3 - 27u + C$$

$$= 9 \sec^3 \theta - 27 \sec \theta + C$$

$$x = 3 \tan \theta$$



$$\tan \theta = \frac{x}{3}$$

$$\sec \theta = \frac{\sqrt{9+x^2}}{3}$$

$$\int \frac{x^3}{\sqrt{x^2+9}} dx = 9 \sec^3 \theta - 27 \sec \theta + C$$

$$= 9 \left(\frac{\sqrt{9+x^2}}{3} \right)^3 - 27 \frac{\sqrt{9+x^2}}{3} + C$$

$$= \left(\frac{\sqrt{9+x^2}}{3} \right)^3 - 9 \sqrt{9+x^2} + C$$

$$3. \int \frac{\sqrt{x^2-1}}{x^6} dx$$

Solution

$$x = \sec \theta$$

$$dx = \tan \theta \sec \theta d\theta$$

$$\sqrt{x^2-1} = \sqrt{\sec^2 \theta - 1} = \sqrt{\tan^2 \theta} = \tan \theta$$

$$\text{Then } \int \frac{\sqrt{x^2-1}}{x^6} dx = \int \frac{\tan \theta}{\sec^6 \theta} \tan \theta \sec \theta d\theta$$

$$= \int \frac{\tan^2 \theta}{\sec^5 \theta} d\theta = \int \frac{\frac{\sin^2 \theta}{\cos^2 \theta}}{\frac{1}{\cos^5 \theta}} d\theta =$$

$$= \int \cos^3 \theta \cdot \sin^2 \theta d\theta = \int \cos^2 \theta \cdot \sin^2 \theta \cos \theta d\theta$$

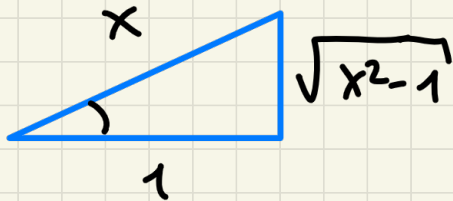
$$= \int (1 - \sin^2 \theta) \sin^2 \theta \cdot \cos \theta d\theta =$$

$$= \int (1 - u^2) u^2 du = \int (u^2 - u^4) du =$$

$$\frac{u^3}{3} - \frac{u^5}{5} + C = \boxed{\frac{\sin^3 \theta}{3} - \frac{\sin^5 \theta}{5} + C}$$

$$\sec \theta = \frac{x}{1}$$

$$\sin \theta = \frac{\sqrt{x^2 - 1}}{x}$$



Then:

$$\int \frac{\sqrt{x^2-1}}{x^6} dx = \frac{\sin^3 \theta}{3} - \frac{\sin^5 \theta}{5} + C$$

$$= \frac{1}{3} \left(\frac{\sqrt{x^2-1}}{x} \right)^3 - \frac{1}{5} \left(\frac{\sqrt{x^2-1}}{x} \right)^5 + C$$

January 21, 2026

Integrals of Rational Functions using Partial Fraction Decomposition

Polynomials: $P(x) = a_0 + a_1x + \dots + a_nx^n$

Rational functions: $f(x) = \frac{P(x)}{Q(x)}$

where $P(x)$, $Q(x)$ are polynomials

$f(x) = \frac{P(x)}{Q(x)}$ is a "proper" rational function if $\deg P(x) < \deg Q(x)$

Ex

$\frac{2x+7}{3x^2-2x+2}$ is proper

$\frac{4x+2x+7}{3x^2-2x+2}$ is not proper
2 → bigger than

$$\int \frac{P(x)}{Q(x)} dx$$

Step # 0 : boil down to a proper rational function

Division with remainder for nums:

If $a, b > 0$ integers, there exist q, r integers such that $a = bq + r$ $0 \leq r < b$

If $P(x)$ and $Q(x)$ are polynomials, there exist $S(x)$ and $R(x)$ such that:

$$P(x) = Q(x) S(x) + R(x)$$

$$\int \frac{P(x)}{Q(x)} dx = \int \frac{Q(x)S(x) + R(x)}{Q(x)} dx =$$

proper $\nearrow$ rat. funct.

$$\int \left(S(x) + \frac{R(x)}{Q(x)} \right) dx = \int S(x) dx + \int \frac{R(x)}{Q(x)} dx$$

Ex. $\int \frac{2x^3 - 5x^2 + 5x + 3}{x-1} dx$

Solution.

$$\begin{array}{r} \textcolor{blue}{-2x^2 - 3x + 2} \\ x-1 \overline{) 2x^3 - 5x^2 + 5x + 3} \\ \underline{2x^3 - 2x^2} \\ -3x^2 + 5x + 3 \\ \underline{-3x^2 + 3x} \\ 2x + 3 \\ \underline{2x - 2} \\ 5 \end{array}$$

$$\begin{array}{l} \textcolor{red}{P(x)} \\ 2x^3 - 5x^2 + 5x + 3 = \\ \textcolor{red}{Q(x)} \quad \textcolor{red}{S(x)} \quad \textcolor{red}{R(x)} \\ = (x-1)(2x^2 - 3x + 2) + 5 \end{array}$$

$$= \int \frac{(x-1)(2x^2 - 3x + 2) + 5}{x-1} dx =$$

$$= \int \left(2x^2 - 3x + 2 + \frac{5}{x-1} \right) dx =$$

$$= \frac{2x^3}{3} - \frac{3x^2}{2} + 2x + 5 \ln|x-1| + C$$

Raised Goal: $\int \frac{P(x)}{Q(x)} dx$ if $\frac{P(x)}{Q(x)}$ proper

Case 1: $Q(x)$ is a product of linear factors

$$Q(x) = (a_1x + b_1)(a_2x + b_2) \dots (a_kx + b_k)$$

There exists constants $A_1, A_2, A_3, \dots$

$\dots A_k$ such that

$$\frac{P(x)}{Q(x)} = \frac{A_1}{a_1x + b_1} + \frac{A_2}{a_2x + b_2} + \dots + \frac{A_k}{a_kx + b_k}$$

Ex. $\frac{P(x)}{Q(x)} = \frac{\text{Smith of degree zero or less}}{(3x-7)(5x+2)(2x+1)} =$

$$\frac{A}{3x-7} + \frac{B}{5x+2} + \frac{C}{2x+1}$$

Ex 1 $\int \frac{x}{x^2+x-2} dx$

Sol.

$$\text{Factor } x^2 + x - 2 = (x+2)(x-1)$$

The form of the PFD is:

$$\frac{x}{x^2 + x - 2} = \frac{A}{x+2} + \frac{B}{x-1}$$

$$\text{Find } \frac{x}{(x+2)(x-1)} = \frac{A}{x+2} + \frac{B}{x-1} \quad | \cdot (x+2)(x-1)$$

A, B

$$x = A(x-1) + B(x+2)$$

$$x = Ax - A + Bx + 2B$$

$$x = (A+B)x + (2B-A)$$

$$\underline{1} \cdot \underline{x+0} = x = \underline{(A+B)}x + \underline{(2B-A)}$$

$$\begin{cases} A+B = 1 \\ 2B-A = 0 \end{cases}$$

$$A = 2B$$

$$2B+B = 1$$

$$3B = 1$$

$$B = \frac{1}{3}$$

$$A = \frac{2}{3}$$

The P.F.D is :

$$\frac{x}{x^2-x-2} = \frac{2/3}{x+2} + \frac{1/3}{x-1}$$

$$\int \frac{x}{x^2-x-2} dx = \int \left(\frac{2/3}{x+2} + \frac{1/3}{x-1} \right) dx$$

$$= \frac{2}{3} \int \frac{1}{x+2} dx + \frac{1}{3} \int \frac{1}{x-1} dx$$

$$= \boxed{\frac{2}{3} \ln|x+2| + \frac{1}{3} \ln|x-1| + c}$$

Shortcut to find A and B

$$x = A(x-1) + B(x+2)$$

$$\text{take } x=1 \quad 1 = A \cdot 0 + B \cdot 3 \Rightarrow B = \frac{1}{3}$$

$$\text{take } x=-2 \quad -2 = A(-2-1) + B \cdot 0 = A = -\frac{2}{3}$$

!!! okay but doesn't always work!!!

Case 2.

Example:

$$\frac{P(x)}{Q(x)} = \frac{\dots\dots}{(5x+1)(2x+7)^3(x-3)^2}$$

$$\frac{A}{(5x+1)} + \frac{B}{(2x+7)} + \frac{C}{(2x+7)^2} + \frac{D}{(2x+7)^3} + \frac{E}{x-3} + \frac{F}{(x-3)^2}$$

$$(a_i x + b_i)^r$$

$$\frac{B_1}{a_i x + b_i} + \frac{B_2}{(a_i x + b_i)^2} + \dots + \frac{B_r}{(a_i x + b_i)^r}$$

Case 3p

$$\frac{P(x)}{Q(x)} = \frac{\dots\dots}{(5x+1)(2x+7)^3(x-3)^2(x^2+2x+7)(2x^2+3x+19)^3}$$

$$\frac{A}{(5x+1)} + \frac{B}{(2x+7)} + \frac{C}{(2x+7)^2} + \frac{D}{(2x+7)^3} + \frac{E}{x-3} + \frac{F}{(x-3)^2}$$

$$+ \frac{6x+H}{x^2+2x+7} + \frac{Ix+J}{2x^2+3x+19} + \frac{Kx+L}{(2x^2+3x+19)^2} + \frac{Mx+N}{(2x^2+3x+19)^3}$$

Ex

$$\int \frac{x+1}{x^4+x^2} dx$$

Sol.

$$\text{Factor } x^4 + x^2 = \underline{x^2} (\underline{x^2 + 1})$$

Form of the P.F.D is

$$\frac{x+1}{x^4+x^2} = \frac{A}{\underline{x}} + \frac{B}{\underline{x^2}} + \frac{Cx+D}{\underline{x^2+1}} \quad | \quad x^2(x^2+1)$$

Find A, B, C, D

$$x+1 = Ax(x^2+1) + B(x^2+1) + (Cx+D)x^2$$

$$x+1 = A(x^3+x) + B(x^2+1) + (Cx^3+Dx^2)$$

$$x+1 = \underline{Ax^3} + \underline{Ax} + \underline{Bx^2} + \underline{B} + \underline{Cx^3} + \underline{Dx^2}$$

$$x+1 = (A+C)x^3 + (B+D)x^2 + Ax + B$$

$$\underline{0} \cdot x^3 + \underline{0} \cdot x^2 + \underline{1} \cdot x + \underline{1} = \underline{(A+C)}x^3 + \underline{(B+D)}x^2 + \underline{A}x + \underline{B}$$

$$\begin{cases} A+C = 0 \\ B+D = 0 \\ A = 1 \\ B = 1 \end{cases} \quad \begin{matrix} C = -1 \\ D = -1 \end{matrix}$$

$$\frac{x+1}{x^4+x^2} = \frac{1}{x} + \frac{1}{x^2} - \frac{x+1}{x^2+1}$$

$$\int \frac{1}{x} + \frac{1}{x^2} - \frac{x+1}{x^2+1} dx =$$

$$= \ln|x| - \frac{1}{x} - \int \frac{x+1}{x^2+1}$$

$$= \int \frac{x}{x^2+1} dx + \underbrace{\int \frac{1}{x^2+1}}_{\tan^{-1} x}$$

$$\text{let } u = x^2+1$$

$$du = 2x dx$$

$$\frac{1}{2} \int \frac{1}{u} du + \tan^{-1} x + C = \frac{1}{2} \ln(x^2+1) + \tan^{-1} x$$

$$= \ln|x| - \frac{1}{x} - \frac{1}{2} \ln(x^2+1) - \tan^{-1} x + C$$